

Group Index of 550 nm TE Silicon Strip Waveguides from Unbalanced MZIs, and Directional-Coupler Split Ratio

edX: mvarenik

Abstract

Design proposal for chip EBeam_mvarenik. 4 Y-branch MZIs with 550×220 nm arms ($\Delta L = 50/75/100/200 \mu\text{m}$) give the TE0 group index (simulated $n_g = 4.140$ at 1550 nm). Five directional couplers ($L_c = 0\text{--}20 \mu\text{m}$) give the split ratio vs. coupling length.

1 Introduction

n_g sets the FSR of rings and interferometers and the length of phase shifters. The DC split ratio sets ring coupling and interferometer balance.

Objectives:

- $n_g(\lambda)$ of TE0, $w = 550$ nm, from the MZI FSR;
- $1/\text{FSR} \propto \Delta L$ over 4 values of ΔL ;
- DC cross-coupling κ^2 vs. L_c and λ .

2 Theory

2.1 Waveguide Compact Model

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (1)$$

2.2 MZI Transfer Function

Two ideal Y-branches [1]:

$$\frac{I_o}{I_i} = \frac{1}{4} \left| e^{-j\beta_1 L_1 - \frac{\alpha}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha}{2} L_2} \right|^2, \quad \beta = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (2)$$

Identical, lossless arms, $\Delta L = L_2 - L_1$:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)]. \quad (3)$$

2.3 Group Index from the FSR

Arm phase difference $\phi = 2\pi n_{\text{eff}} \Delta L / \lambda$. Adjacent minima differ by $\Delta\phi = 2\pi$, so $|d\phi/d\lambda| \text{FSR} = 2\pi$ with

$$\frac{d\phi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2} \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) \Rightarrow \underset{\tilde{n}_g}{n_g} = \frac{\lambda^2}{\Delta L \cdot \text{FSR}}. \quad (4)$$

For several ΔL :

$$\frac{1}{\text{FSR}} = \frac{n_g}{\lambda^2} \Delta L, \quad \frac{\delta n_g}{n_g} \approx \sqrt{2} \frac{\delta \lambda_{\text{min}}}{\text{FSR}}. \quad (5)$$

2.4 Directional Coupler

$$\kappa^2 = \frac{P_{\text{cross}}}{P_{\text{cross}} + P_{\text{thru}}} = \sin^2 \left(\frac{\pi(L_c + L_0)}{2L_x} \right) \quad (6)$$

L_x : cross-over length; L_0 : coupling in the bend regions [1].

3 Modelling and Simulation

3.1 Waveguide

Si strip, $w = 550$ nm, $h = 220$ nm, SiO_2 cladding, TE polarization. MZI arms are 550 nm wide, with identical 500→550 nm tapers in both arms. Lumerical MODE (FDE) [2]: Figs. 1, 2.

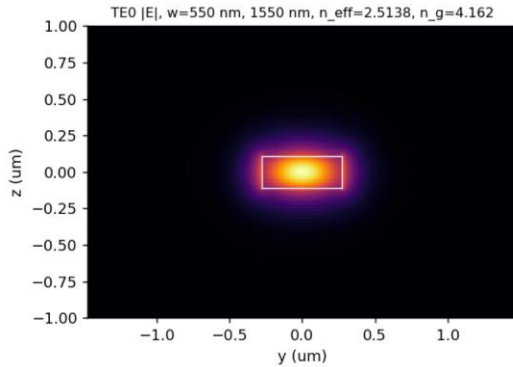


Fig. 1. TE0 mode profile ($|E|$) at 1550 nm, MODE (FDE).

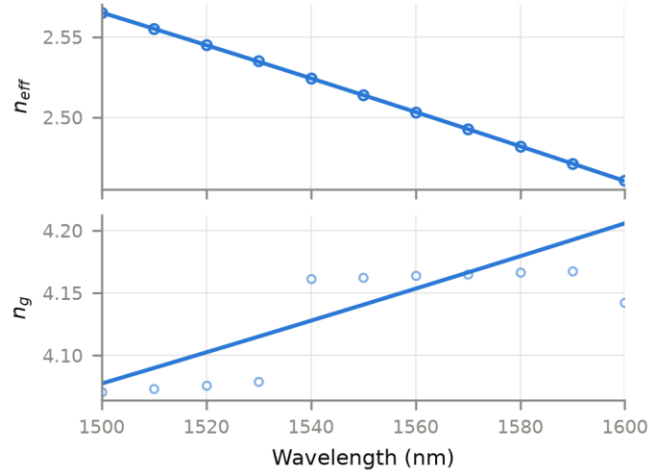


Fig. 2. n_{eff} and n_g vs. wavelength. Markers: MODE (FDE); lines: Eq. (7).

Compact model, λ in μm , $\lambda_0 = 1.55 \mu\text{m}$:

$$n_{\text{eff}}(\lambda) = 2.5138 - 1.0495(\lambda - \lambda_0) - 0.4139(\lambda - \lambda_0)^2. \quad (7)$$

$n_g = 4.077 / 4.140 / 4.206$ at 1500 / 1550 / 1600 nm.

3.2 MZI Design

TABLE 1. Expected FSR from Eq. (7)

Device	ΔL (μm)	FSR (nm) at $\lambda =$			pts/ FSR
		1500	1550	1600	
MZITEw550dL50	50	11.04	11.61	12.17	11037
MZITEw550dL75	75	7.36	7.74	8.12	7358
MZITEw550dL100	100	5.52	5.80	6.09	5518
MZITEw550dL200	200	2.76	2.90	3.04	2759

ebeam_y_1550 split and combine. FSR < 50 nm and > 2700 points per FSR at 1 pm (Table 1).

3.3 Circuit Simulation

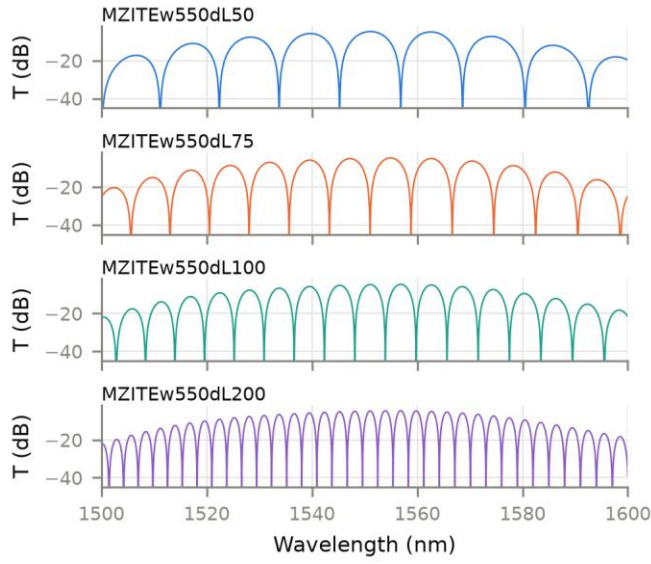


Fig. 3. Simulated MZI transmission, GC to GC: Eq. (3) with Eq. (7) and the PDK TE GC response [3].

Fig. 3.

3.4 Extraction Test on Simulated Spectra

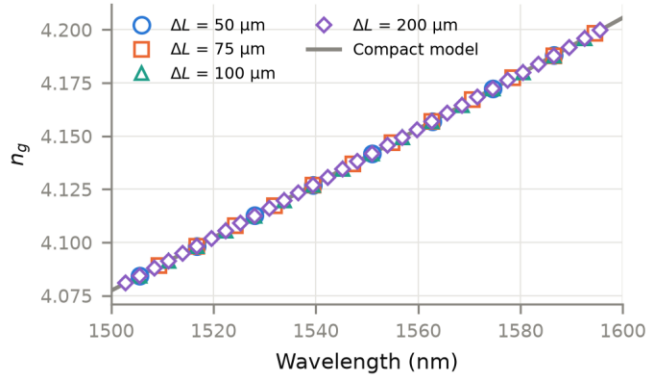


Fig. 4. n_g from the simulated FSR, Eq. (4) (markers), and Eq. (7) (line).

TABLE 2. n_g at 1550 nm extracted from Fig. 3

Device	FSR (nm)	n_g
MZITEw550dL50	11.605	4.1405
MZITEw550dL75	7.737	4.1405
MZITEw550dL100	5.802	4.1405
MZITEw550dL200	2.901	4.1405
Slope, Eq. (5), MZITEw550		4.1404

Eqs. (4), (5) recover the model n_g (Fig. 4, Table 2).

3.5 Procedure for Measured Data

1. Subtract the LBTE spectrum (dB).
2. Locate the minima; quadratic fit around each null (sub-pm).
3. FSR of adjacent minima $\rightarrow n_g(\lambda)$, Eq. (4).
4. Slope over ΔL , Eq. (5); compare with Eq. (7).

3.6 Directional Couplers

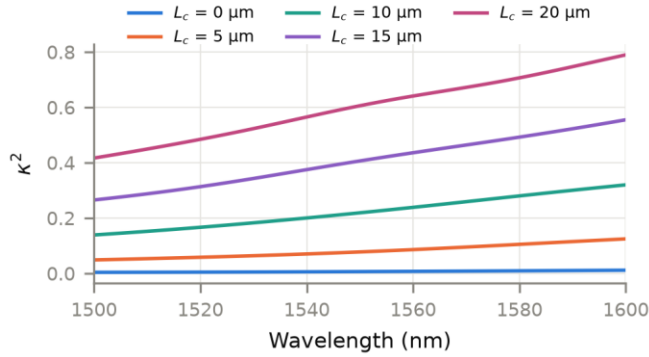


Fig. 5. Expected DC cross-coupling vs. wavelength, PDK ebeam_dc_te1550 models [3], $g = 200$ nm.

TABLE 3. Expected DC cross-coupling at 1550 nm

Device	L_c (μm)	κ^2 (PDK)	κ^2 (fit)
DCL0	0	0.006	0.006
DCL5	5	0.078	0.077
DCL10	10	0.219	0.215
DCL15	15	0.406	0.398
DCL20	20	0.606	0.598

ebeam_dc_te1550, $g = 200$ nm, 500 nm waveguides; 1 input, both outputs to GCs, unused input terminated. Fit of Eq. (6): $L_x = 39.1 \mu m$, $L_0 = 2.0 \mu m$ (Fig. 5, Table 3). Measured: both outputs use the same GC type, so the GC loss cancels in κ^2 ; fit Eq. (6) at each λ .

3.7 Loopback

$IL_{GC} = 1/2 T_{LB}$ [dB]. Expected (PDK TE GC model [3]): LBTE peak -4.6 dB at 1555 nm, 3 dB bandwidth 47 nm.

4 Fabrication

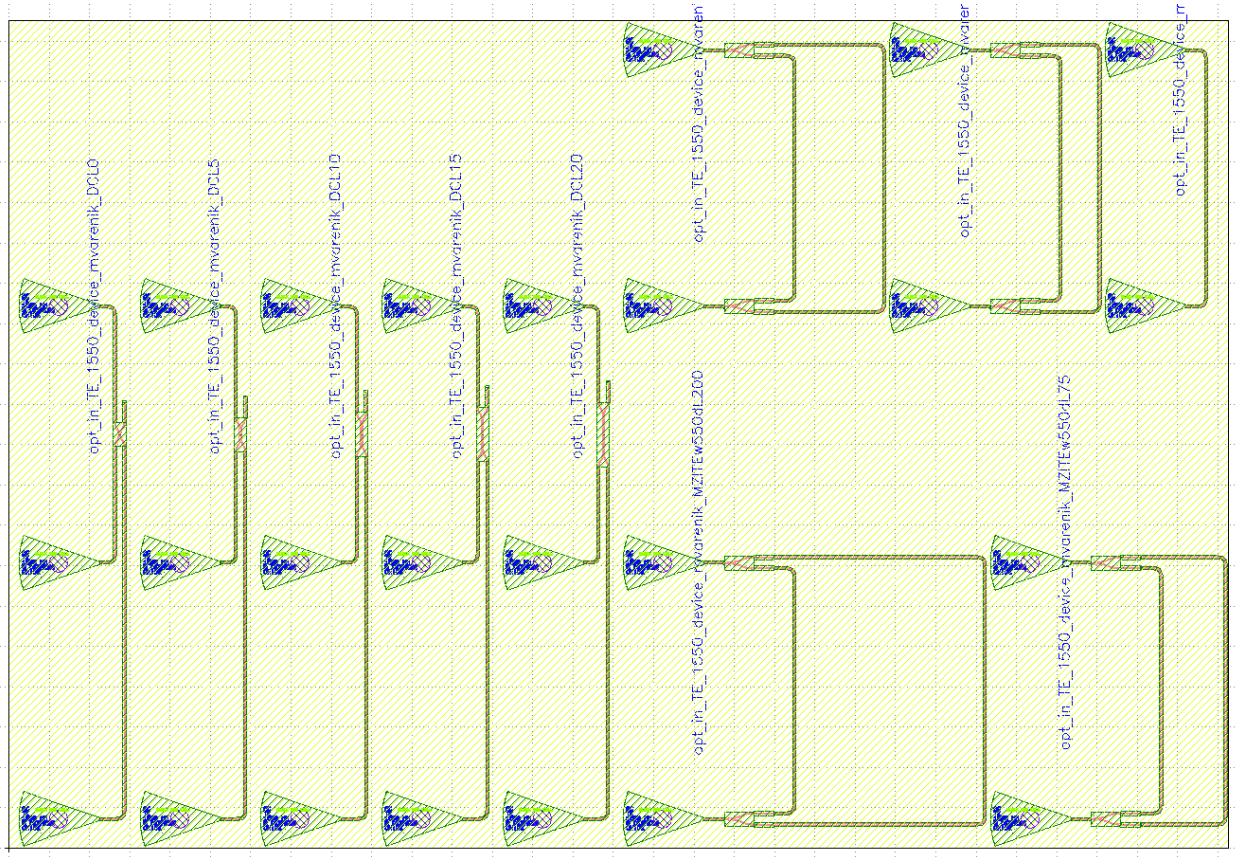


Fig. 6. Layout of EBeam_mvarenik ($605 \times 410 \mu\text{m}$).

TABLE 4. Devices on EBeam_mvarenik

Device	Pol.	Description
DCL0	TE	DC, $g=200 \text{ nm}$, $L_c=0 \mu\text{m}$
DCL5	TE	DC, $g=200 \text{ nm}$, $L_c=5 \mu\text{m}$
DCL10	TE	DC, $g=200 \text{ nm}$, $L_c=10 \mu\text{m}$
DCL15	TE	DC, $g=200 \text{ nm}$, $L_c=15 \mu\text{m}$
DCL20	TE	DC, $g=200 \text{ nm}$, $L_c=20 \mu\text{m}$
MZITew550dL200	TE	MZI, Y-branch, $\Delta L=200 \mu\text{m}$, $w=550 \text{ nm}$
MZITew550dL75	TE	MZI, Y-branch, $\Delta L=75 \mu\text{m}$, $w=550 \text{ nm}$
MZITew550dL100	TE	MZI, Y-branch, $\Delta L=100 \mu\text{m}$, $w=550 \text{ nm}$
MZITew550dL50	TE	MZI, Y-branch, $\Delta L=50 \mu\text{m}$, $w=550 \text{ nm}$
LBTE	TE	GC loopback, $L=144 \mu\text{m}$

KLayout, SiEPIC-Tools [4], EBeam PDK [3] (Fig. 6, Table 4). TE GCs, 127 μm pitch; labels `opt_in_TE_1550_device_mvarenik_<name>`. Functional check: 0 errors; DRC: 0 errors. Electron-beam lithography, Applied Nanotools NanoSOI [5], SiEPIC openEBL.

5 Experimental Data

To be completed (1500–1600 nm sweep, 1 pm step).

6 Analysis

To be completed.

7 Conclusion

To be completed.

References

- [1] L. Chrostowski and M. Hochberg, *Silicon photonics design: From devices to systems*. Cambridge, UK: Cambridge University Press, 2015.
- [2] Ansys Inc., “Ansys Lumerical MODE, FDTD and INTERCONNECT, 2026 R1.” <https://www.ansys.com/products/optics>, 2026.
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- [4] SiEPIC, “SiEPIC-Tools for KLayout, version 0.5.31.” <https://github.com/SiEPIC/SiEPIC-Tools>, 2026.
- [5] Applied Nanotools Inc., “NanoSOI electron-beam lithography fabrication process.” <https://www.appliednt.com/nanosoi-fabrication-service/>, 2026.